
Complements of Operational Research

Exercise sheet 1 — Decision theory

Exercises marked **homework** are to be solved at home and sent by email to the lecturer, by the deadline given in class; the solution is shared afterwards. The others are from previous tests and exams. In all matrices, the rows are the actions and the columns the states of nature.

1.2 Non-probabilistic decision criteria

Exercise 1 — The students' union game

homework

The students' union organises the following game: 30 white balls, 20 black balls and 10 red balls are placed in a black bag; the player chooses game A or game B and draws a ball at random. Gains (€):

Decision	White	Black	Red
Game A	−20	−5	100
Game B	−5	5	50

Which game do you decide to play, according to the criteria:

- a) Maximax; b) Maximin; c) Laplace; d) Savage; e) Hurwicz with coefficient of optimism 0.7?
- f) (After section 1.3.) Use the contents of the bag to assign a probability to each colour. Which game do you choose by the expected value criterion? Compare with c).

Exercise 2 — Maintenance plan for a fleet

homework

A transport company is choosing the maintenance plan for its fleet for next year. The total annual **cost** (maintenance plus repairs, in thousands of euros) depends on the number of breakdowns that occur:

Plan	few breakdowns	moderate breakdowns	many breakdowns
corrective (repair only)	10	40	95
quarterly preventive	25	35	60
monthly preventive	45	45	50
external contract	35	40	65

- a) Is any plan dominated?
- b) Which plan should be chosen according to the optimistic, pessimistic, Laplace, Savage and Hurwicz (coefficient of optimism 0.6) criteria?

Exercise 3 — How much to order?

homework

A shop has to order q units of a seasonal product *before* knowing the season's demand d . Each unit costs 6 € and sells for 10 €; units that are not sold are cleared at the end of the season at 3 € each. The shop assumes that demand will be 100, 200, 300 or 400 units, and only orders in batches of 100.

- a) Build the decision matrix: actions $q \in \{100, 200, 300, 400\}$, states $d \in \{100, 200, 300, 400\}$, consequences: profit for the season (€).
- b) How much should be ordered according to the Maximax, Maximin, Laplace, Savage and Hurwicz criteria with coefficient of optimism 0.7?
- c) What attitude towards risk corresponds to each of the quantities chosen?

1.3–1.4 Probabilistic criteria and subjective probabilities

Exercise 4 — Ordering umbrellas

homework

A shop has to decide how many umbrellas to order for the season. The result (m.u.) depends on the demand:

Order	strong demand	moderate demand	weak demand
nothing	−40	0	70
40	−10	15	35
80	50	25	−30
160	80	20	−35

Using the criteria indicated, determine which decision to take:

- a) Maximax; b) Maximin; c) Laplace; d) Hurwicz with coefficient of optimism 0.7; e) Savage;
- f) maximisation of the expected value, with probabilities 30%, 40% and 30%;
- g) minimisation of the expected opportunity loss, with the same probabilities.

Exercise 5 — Three decisions, three states

1st test, 13/04/2018

Consider the following table of estimated gains (m.u.) for 3 possible decisions (D_1 , D_2 , D_3), depending on which of the 3 states of nature A_1 , A_2 , A_3 occurs:

Decision	A_1	A_2	A_3
D_1	50	30	20
D_2	100	50	−10
D_3	350	−20	−60

- a) Which action should the company take with the criteria of (i) Laplace; (ii) Hurwicz with a coefficient of optimism of 0.70?
- b) Considering that the market shows probabilities of occurrence of 30%, 50% and 20% for states A_1 , A_2 and A_3 , respectively, determine the decision to take with the criterion of minimisation of the expected opportunity loss.

Exercise 6 — Sales staff in a footwear company

1st test, 08/04/2019

A footwear company has to take one of the following actions: A_1 — increase the sales staff by 10%; A_2 — keep the sales staff; A_3 — reduce the sales staff by 10%. The result depends on the economic situation: C_1 — good; C_2 — mediocre; C_3 — bad. Estimated gains (m.u.):

Actions	C_1	C_2	C_3
A_1	300	−100	−200
A_2	100	50	−50
A_3	50	20	0

- a) Which action should the company take with the criteria of (i) Laplace; (ii) Hurwicz with a coefficient of optimism of 0.75?
- b) The decision maker was asked for the relative odds between the 3 states of nature, and answers: “The relative odds of state C_1 with respect to C_2 are 3 to 1” and “The relative odds of state C_2 with respect to C_3 are 2 to 1”. Using the two answers, determine the decision to take with the criterion of minimisation of the expected opportunity loss.

Exercise 7 — Four actions, three states

exam, 05/07/2021

Consider the following table of estimated gains (m.u.) with actions $A_1, \dots, A_4$ and states of nature C_1, C_2, C_3 :

Actions	C_1	C_2	C_3
A_1	-40	0	70
A_2	-15	30	35
A_3	60	45	-20
A_4	10	30	15

- Which action should the company take with the criteria of (i) Laplace; (ii) Hurwicz with a coefficient of optimism of 0.75?
- Considering that the market shows probabilities of 35%, 40% and 25% for C_1, C_2 and C_3 , determine the action to take with the criterion of minimisation of the expected opportunity loss.

Exercise 8 — Four decisions and subjective probabilities (I)

1st test, 03/04/2024

Decision	A_1	A_2	A_3
D_1	-38	10	45
D_2	-9	24	42
D_3	41	17	-25
D_4	66	14	-1

- Which action should the company take with the criteria of (i) Laplace; (ii) Hurwicz with a coefficient of optimism of 0.60?
- The decision maker answers: "The relative odds of state A_1 with respect to A_2 are 1 to 2" and "The relative odds of state A_2 with respect to A_3 are 1 to 3". Determine the decision to take with the criterion of minimisation of the expected opportunity loss.

Exercise 9 — Four decisions and subjective probabilities (II)

1st test, 27/03/2025

Decision	A_1	A_2	A_3
D_1	-28	5	15
D_2	9	14	2
D_3	49	17	-5
D_4	42	10	-1

- Which action should the company take with the criteria of (i) Laplace; (ii) Hurwicz with a coefficient of optimism of 0.70?
- The decision maker answers: "The relative odds of state A_1 with respect to A_2 are 1 to 3" and "The relative odds of state A_2 with respect to A_3 are 1 to 2". Determine the decision to take with the criterion of minimisation of the expected opportunity loss.